1.1 Points, Lines, and Planes

Point, line, and plane are the three undefined terms that every later definition and postulate is built from. Beyond simply naming a ray or a segment, this section pushes into the logic behind collinearity and coplanarity — using postulates to justify why three points determine a plane only when they are noncollinear, and why two planes can only intersect in a line, never in a single stray point. Expect problems that combine several undefined terms in one figure, ask you to name the postulate that justifies an answer, and require reasoning about three-dimensional configurations rather than just flat sketches.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Name the line through M and N.

2. Write the ray that starts at A and goes through B.

3. Write the segment with endpoints P and Q.

4. Points A, B, C do not lie on one line. Are they collinear?

5. Three noncollinear points A, B, C. Are they coplanar?

6. Points R, S, T collinear with S between R and T. Name a pair of opposite rays.

7. How many planes contain three given noncollinear points?

8. How many lines contain two given distinct points?

9. If planes P and Q intersect (and are not equal), what shape is $\mathrm{P} \cap \mathrm{Q}$?

10. Does $\overline{AB}$ have finite length?

11. Does $\overrightarrow{AB}$ have finite length?

12. Line ℓ and point $G \notin \ell$ are given. How many planes contain both?

13. Lines j and k intersect at P; $Q \in j$ and $R \in k$ with $Q, R \neq P$. Name the plane containing j and k.

14. Five points are positioned so that no 3 are collinear and no 4 are coplanar. How many planes are determined by choosing 3 points at a time?

15. A tetrahedron has 4 vertices, no 3 collinear and not all 4 coplanar. Using combinations, find the number of edges (lines) and the number of face planes, and explain the pattern.

✓ Answer Key

1. $\overleftrightarrow{MN}$
2. $\overrightarrow{AB}$
3. $\overline{PQ}$
4. No
5. Yes — any 3 points, collinear or not, lie in at least one common plane
6. $\overrightarrow{SR}$ and $\overrightarrow{ST}$
7. Exactly one
8. Exactly one
9. A line
10. Yes
11. No — it extends forever past B
12. Exactly one
13. Plane PQR (or QPR, etc.)
14. $\binom{5}{3} = 10$ planes
15. $\binom{4}{2} = 6$ edges; $\binom{4}{3} = 4$ face planes — edges come from pairs of vertices, faces from triples